claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a29cb1f5ed13d6c29.jsonl`
- SHA-256: `8dde02f8164a52ee4c047791eeab32e463b0a20fab43ba1c413423e47326809e`
- Source modified: `2026-05-31T04:20:29+00:00`
- Imported at: `2026-07-05T16:48:20+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:18:45.260Z)

Source Extractor

```
Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision /
document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud
API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date,
narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST
personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-
column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both
born-digital (text layer present) and the harder cases where pdfplumber's
extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies
between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default.
Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud
options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT,
BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-
commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This
is critical ? many document-AI models and their training datasets are non-commercial (e.g.
LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several
VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo
license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24
GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license
(state explicitly), latest release / maintenance status as of 2025-2026, VRAM need +
quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL
TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL
(note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 /
2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community
License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR,
EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND
PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),
MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON
```

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/2508.05201
Title: FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance (ICAIF '25)
Found via: Borderless-table accuracy & financial-number hallucination/verification (contrarian/skeptical) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:18:48.712Z)

I'll fetch the source and extract key claims relevant to the research question about local document-AI/OCR models for parsing bank statement PDFs.

3. user (2026-05-31T04:19:02.506Z)

Key Findings on Intrinsic Tabular Hallucinations in Finance

Publication Details:
- **Title:** FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance
- **Authors:** Mengao Zhang, Jiayu Fu, Tanya Warrier, Yuwen Wang, Tianhui Tan, Ke-wei Huang
- **Venue:** 6th ACM International Conference on AI in Finance (ICAIF '25), November 15-18, 2025, Singapore
- **Date:** October 27, 2025

---

(1) Problem Definition: Intrinsic Tabular Hallucinations

The research addresses how large language models and vision-language models produce erroneous financial data when processing tables. The framework specifically focuses on understanding how models misrepresent numerical information in financial contexts without relying on external knowledge beyond training data.

---

(2) Types of Hallucinations on Financial Numbers

While the PDF metadata and structure indicate the paper examines how models hallucinate on financial numbers in tables, the specific truncated content prevents detailed extraction of explicit hallucination categories (transposed digits, dropped rows, duplicated rows, wrong values). The document references prior work on hallucination detection but the main findings section is in the compressed content streams.

---

(3) Quantitative Results on Hallucination Rates

The document structure indicates Tables 8, 9, 10, and 13 contain quantitative results, but the specific accuracy metrics and hallucination rates across models are contained in the compressed content streams that were truncated in the source material provided.

---

(4) Models Tested and Comparisons

The paper mentions testing multiple models, with references suggesting evaluation across different LLM and VLM architectures. However, the specific model names, versions, and comparative performance data are located in sections not fully readable from the truncated PDF metadata.

---

(5) FAITH Framework for Detection/Measurement

The framework title indicates a structured approach to "assessing intrinsic tabular hallucinations," suggesting a methodology for measurement. Keywords indicate focus on "benchmarking" and "model verification and validation," but detailed framework specifications are in inaccessible sections.

---

(6) Implications for Financial Table Verification

The research's positioning within finance applications and emphasis on hallucination assessment suggests implications for critical financial analysis, but specific recommendations are located in truncated sections.

Note: The PDF content provided was heavily compressed with FlateDecode streams. A readable version would better support extraction of the quantitative results, specific models evaluated, and detailed framework specifications.

[Binary content (application/pdf, 1006.3KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\tool-results\webfetch-1780201019681-c2bd5q.pdf]

4. assistant (2026-05-31T04:19:05.743Z)

The PDF content was compressed and the WebFetch couldn't extract the substantive findings. The PDF was saved locally. Let me read it directly to extract the actual claims. I'll also try the HTML abstract page in parallel for context.

5. user (2026-05-31T04:19:06.766Z)

```
pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)
```

6. user (2026-05-31T04:19:17.607Z)

FAITH Framework Paper Summary

Abstract (Verbatim)

```
"Hallucination remains a critical challenge for deploying Large Language Models (LLMs) in
finance. Accurate extraction and precise calculation from tabular data are essential for
reliable financial analysis, since even minor numerical errors can undermine decision-making and
regulatory compliance. Financial applications have unique requirements, often relying on
context-dependent, numerical, and proprietary tabular data that existing hallucination
benchmarks rarely capture. In this study, we develop a rigorous and scalable framework for
evaluating intrinsic hallucinations in financial LLMs, conceptualized as a context-aware masked
span prediction task over real-world financial documents. Our main contributions are: (1) a
novel, automated dataset creation paradigm using a masking strategy; (2) a new hallucination
evaluation dataset derived from S&P 500 annual reports; and (3) a comprehensive evaluation of
intrinsic hallucination patterns in state-of-the-art LLMs on financial tabular data. Our work
provides a robust methodology for in-house LLM evaluation and serves as a critical step toward
building more trustworthy and reliable financial Generative AI systems."
```

Key Metadata

```
- **Venue:** AMC ICAIF'25
- **Submission Date:** August 7, 2025 (revised October 24, 2025)
- **Authors:** Mengao Zhang, Jiayu Fu, Tanya Warrier, Yuwen Wang, Tianhui Tan, Ke-wei Huang
- **Classification:** Machine Learning (cs.LG)
```

Notable Findings

```
The document describes the FAITH framework as implementing "a context-aware masked span
prediction task" using real financial documents. However, the provided content contains no
specific hallucination rates, detailed model comparisons, or empirical results?only the
conceptual framework description.
```

7. assistant (2026-05-31T04:19:20.754Z)

```
The abstract gives me the core framing. The PDF needs a tool to extract text. Let me try Python
(pdfplumber/pypdf are likely available given the project context) to extract the key results
sections.
```

8. user (2026-05-31T04:19:22.239Z)

```
no pypdf No module named 'pypdf'
no pdfplumber No module named 'pdfplumber'
```

9. assistant (2026-05-31T04:19:24.622Z)

```
Not in the global env. The project uses these though. Let me find the project's Python
environment.
```

10. user (2026-05-31T04:19:26.125Z)

```
Exit code 2
total 77
drwxr-xr-x 1 avidu 197609     0 May 31 08:52 .
drwxr-xr-x 1 avidu 197609     0 May 30 21:47 ..
drwxr-xr-x 1 avidu 197609     0 May 31 09:42 .git
-rw-r--r-- 1 avidu 197609   286 May 30 23:27 .gitattributes
drwxr-xr-x 1 avidu 197609     0 May 31 08:52 .githooks
drwxr-xr-x 1 avidu 197609     0 May 30 23:27 .github
-rw-r--r-- 1 avidu 197609  1403 May 31 08:39 .gitignore
```

```
drwxr-xr-x 1 avidu 197609     0 May 30 22:57 .pytest_cache
drwxr-xr-x 1 avidu 197609     0 May 30 22:57 .ruff_cache
drwxr-xr-x 1 avidu 197609     0 May 30 22:57 .venv
-rw-r--r-- 1 avidu 197609  4273 May 30 21:54 CLAUDE.md
-rw-r--r-- 1 avidu 197609 17507 May 30 23:43 DESIGN.md
-rw-r--r-- 1 avidu 197609  1080 May 31 08:52 Makefile
-rw-r--r-- 1 avidu 197609  2925 May 31 00:03 README.md
drwxr-xr-x 1 avidu 197609     0 May 31 09:42 docs
-rw-r--r-- 1 avidu 197609  1916 May 31 08:39 pyproject.toml
drwxr-xr-x 1 avidu 197609     0 May 30 22:57 src
drwxr-xr-x 1 avidu 197609     0 May 31 08:37 testdata
drwxr-xr-x 1 avidu 197609     0 May 31 08:52 tests
drwxr-xr-x 1 avidu 197609     0 May 31 08:39 tools
---VENV---
.venv/Scripts/python.exe
---POETRY---
```

11. assistant (2026-05-31T04:19:29.222Z)

There's a `.venv`. Let me use the project's Python to extract the PDF text.

12. user (2026-05-31T04:19:33.666Z)

```
PAGES 9
FAITH: A Framework for Assessing Intrinsic Tabular
Hallucinations in Finance
MengaoZhang? JiayuFu* TanyaWarrier*
mengaoz@nus.edu.sg jiayu.fu@u.nus.edu tanya.warrier@u.nus.edu
AsianInstituteofDigitalFinance, AsianInstituteofDigitalFinance, AsianInstituteofDigitalFinance,
NationalUniversityofSingapore NationalUniversityofSingapore NationalUniversityofSingapore
Singapore Singapore Singapore
YuwenWang TianhuiTan Ke-WeiHuang
wangyuwen@u.nus.edu tant@nus.edu.sg huangkw@comp.nus.edu.sg
AsianInstituteofDigitalFinance, AsianInstituteofDigitalFinance, AsianInstituteofDigitalFinance,
NationalUniversityofSingapore NationalUniversityofSingapore NationalUniversityofSingapore
Singapore Singapore Singapore
Abstract AIinFinance(ICAIF?25),November15?18,2025,Singapore,Singapore.ACM,
HallucinationremainsacriticalchallengefordeployingLargeLan-
NewYork,NY,USA,9pages.https://doi.org/10.1145/3768292.3770433
guageModels(LLMs)infinance.Accurateextractionandprecisecal-
culationfromtabulardataareessentialforreliablefinancialanalysis, 1 Introduction
sinceevenminornumericalerrorscanunderminedecision-making
WhileLargeLanguageModels(LLMs)arerapidlytransformingfi-
andregulatorycompliance.Financialapplicationshaveuniquere-
nancialservicesthroughcapabilitiessuchasautomatedinformation
quirements,oftenrelyingoncontext-dependent,numerical,and
extraction[4]andclient-facingchatbots[16],theirdeploymentalso
proprietarytabulardatathatexistinghallucinationbenchmarks
introducessubstantialrisks,amongwhichhallucinationposesseri-
rarelycapture.Inthisstudy,wedeveloparigorousandscalable
ousthreatstodecision-makingandstakeholdertrust[8,19?22].In
frameworkforevaluatingintrinsichallucinationsinfinancialLLMs,
response,regulatorssuchastheMonetaryAuthorityofSingapore
conceptualizedasacontext-awaremaskedspanpredictiontaskover
(MAS)haveunderscoredtheneedforrobustmodelriskmanage-
real-worldfinancialdocuments.Ourmaincontributionsare:(1)a
mentframeworkstailoredtoadvancedAIapplications[15].Despite
novel,automateddatasetcreationparadigmusingamaskingstrat-
growingregulatoryscrutiny,systematicmethodsforevaluating
egy;(2)anewhallucinationevaluationdatasetderivedfromS&P
andmitigatinghallucinationsinfinancialcontextsremainlargely
500annualreports;1and(3)acomprehensiveevaluationofintrinsic
undeveloped.Asafoundationalstep,thisstudyproposesascal-
hallucinationpatternsinstate-of-the-artLLMsonfinancialtabu-
able,comprehensiveframeworkforevaluatingandanalyzingLLM
```

lardata.Ourworkprovidesarobustmethodologyforin-house
hallucinationsinfinance,quantifieshowincreasinginformation-
LLMevaluationandservesasacriticalsteptowardbuildingmore
extractioncomplexityexacerbateshallucinationerrors,andoffers
trustworthyandreliablefinancialGenerativeAIsystems.
actionableguidanceforresearchersandpractitioners.
Recentempiricalstudieshavebeguntorevealthathallucinations
CCSConcepts
infinanceexhibitdomain-specificcharacteristicscomparedwith
?Computingmethodologies?Naturallanguageprocessing; thoseobservedingeneral-
domainNLPbenchmarks.Forinstance,
Modelverificationandvalidation;?Appliedcomputing?
Shahetal.[22]findthatLLMs?factualaccuracyonhistoricaldata
Economics;?Informationsystems?Informationretrieval.
variesacrosscompanysizeandfinancialperiods,whileKangand
Liu[8]andSarmahetal.[21]demonstratethatevenstate-of-the-
Keywords artmodelsfrequentlymisinterpretfinancialtermsormisreport
LargeLanguageModels,Hallucination,Benchmarking,Financial
numericalvalues.ComplementaryworksuchasRoychowdhury
NLP,TableReasoning [19]outlinesstagestowardhalluci

13. assistant (2026-05-31T04:19:37.807Z)

Good, the text extraction works. Now let me extract the key results sections ? particularly the findings on how complexity increases hallucination, and any quantitative results.

14. user (2026-05-31T04:19:41.017Z)

nsichallucina-
Yet,evaluatinghallucinationsinasystematicandrobustman- tionpatternsinstate-of-the-
artLLMsonfinancialtabular
nerremainsanon-trivialchallengeduetothescarcityoffinance-
data,includingadetailedbreakdownbyreasoningtype.
specificbenchmarks.Existingdatasetsarepredominantlyderived
fromgeneral-domaintextssuchasWikipedia[3,6],whereasfinan- 2 RelatedWorks
cialapplicationsinvolvenumericallygrounded,context-dependent
2.1 HallucinationEvaluationDatasets
dataandspecializedterminology.Moreimportantly,thereisaclear
needforanautomatedandscalableapproachtocreatingevalu- LLM hallucinations can be categorized as extrinsic or intrinsic,
ationdatasetsforhallucinationdetectioninreal-worldfinancial
basedonthereferencesource[3].Theevaluationofextrinsichal-
settings.Manualannotationisresource-intensiveandcanhardly lucinations assesses whether a model?s output, generated from
keepupwithevolvingLLMbehaviorsorthediversityofproprietary
itsinternalknowledgealone,alignswithestablishedworldfacts.
financialdata. Benchmarksinthiscategory[10,11,13]generallyevaluatemod-
Toaddressthesegaps,wedevelopFAITH(AFrameworkfor elsagainststatic,open-
domaincorporalikeWikipedia.Financial
AssessingIntrinsicTabularHallucinationsinFinance)-anauto-
LLMs,however,operateinadomainwhereinformationishighly
matedandscalableapproachforconstructingintrinsichallucination time-sensitive,context-
intensive,andoftenproprietary,requiring
evaluationtasksdirectlyfromrealfinancialannualreports,where
evaluationmethodsthatprioritizecontextualaccuracyoverpre-
ground-truthnumericalvaluesareexplicitlyavailable.Weaddressa
trained,parametricknowledge[8,19?22].
corechallengeinfinancialLLMapplications:accuratelyidentifying,
Therefore,intrinsichallucinationevaluationismorerelevantand
calculating,andpresentingnumericalvaluesbasedonfinancial
criticalforfinancialapplications,asitdirectlymeasuresamodel?s
statements(inputcontext).Weintroduceanovelcontext-aware
faithfulnesstoaspecific,providedcontext.Whileintrinsicevalu-
maskedspanpredictiontask,wherenumericalvaluesfromreal
ationdatasetshaveemerged,especiallyforRetrieval-Augmented

annualreportsaremaskedandLLMsarepromptedtorecoverthem.
Generation(RAG)applications[6,14,17],thosebasedongeneral-
Toensurereliableevaluation,wedevelopanovelfilteringmethod
purposetext,mostlyarelimitedtosimplelook-uptasksratherthan
thatselectsonlythosevalueswithaunique,consistent,andan- complexreasoning[2,17].
swerablegroundtruth.Acentralcontributionisourtaxonomyof
Ourworkaddressesthiscriticalgap,focusingonintrinsichallu-
fourmutuallyexclusiveandexhaustivefinancialreasoningtypes:
cinationswithincomplexfinancialtabulardata?auniqueandhigh-
DirectLookup,ComparativeCalculation(Thevalueisafunctionof
stakesdomainlargelyoverlookedbycurrentevaluationmethod-
thesamevariableindifferentperiods,e.g.,year-over-yeargrowth), ologies.
BivariateCalculation(thevalueisafunctionoftwovariables,e.g.,
2.2 TabularReasoningandEvaluation
grossmargin),andMultivariateCalculation(morecomplexdepen-
dencies).Thisclassificationenablesstructuredevaluationacross
Ourfocusonintrinsichallucinationswithtablesconnectstothe
reasoningcomplexity.Usingourapproach,webuildalarge-scale
broaderresearchareaoftabularreasoning.Foundationalbench-
benchmarkdatasetfromS&P500annualreportsandevaluatelead-
markssuchasRealHiTBench[24]andTableBench[25]havebeen
ingLLMsontheirintrinsichallucinationrates.
pivotalinassessingcoreLLMskillslikenumericalreasoning.In
Empirically, our results reveal a stratified hierarchy of
themoreintricatefinancialdomain,specializedbenchmarkshave
modelreliabilityinfinancialreasoning.Proprietaryfrontier
emergedtoaddresshighercomplexity.DatasetslikeFinQA[5]and
modelssuchasClaude-Sonnet-4andGemini-2.5-Proachievehigh TAT-
QA[30]testjointreasoningovertablesandtext,whileothers
overallaccuracy,yeteventhesesystemsexhibit10?20%errorrates
focusonnavigatingcomplexhierarchicalstructures[9,23,27].
onmulti-stepnumericalreasoning.Incontrast,smalleropen-source
WhilethesebenchmarkshavesignificantlyadvancedLLMrea-
modelsoftenfailcatastrophicallyoncomplexcalculations,scoring
soningcapabilities,theirevaluationscopeisprimarilyfocusedon
nearzeroinmultivariatescenarios.Thisdegradationwithreason-
thecorrectnessofthefinalanswer.Thisfocusonaccuracycreates
ingcomplexityhighlightsanurgentlimitationofcurrentarchi-
acriticalblindspot:astaskcomplexityincreases,sodoestherisk
tectures?thedifficultyofmaintainingfactualconsistencyacross thatamodelgeneratesacorrect-
seemingoutputthroughunfaithful
reasoningorhallucination.Addressingthisgaprequiresmoving
beyondtraditionalquestion-answeringformatstodevelopscalable
===PAGE 3===

<LLM Answerability Annotation> <Format Input>
Text before: Services and other revenue consists of ?
Text before: Services and other revenue?
Sentence: Automotive sales revenue increased
<SPAN>$11.30 billion<SPAN>, or <SPAN>17%<SPAN>, Span1: Automotive sales revenue increased
in the year ended December 31, 2023? SpanSplit <MASK> or 17% ?
Span2: ?increased $11.30 billion or
Text after: The increase was partially offset by a lower
<MASK> in the year ended December 31?
average? For each highlighted span, determine
<SentenceExtraction and Numerical Span Identification> if the span is answerable? T b e y x a t
a lo ft w er e : r T av h e e r i a n g c e r ? ease was partially offset
Span1 Span2 <Ground Truth>
Text Before: The following table disaggregates our Span1 Span2
revenue? $11.30 billion 17%
10-K
Reports <LLM Fill in the Blank>
From SEC
Your task is to fill in the masked blank in a

sentence using the provided financial data tables. $11.30 million 16.78%
<Context Extraction> <Prediction>
Figure1:Illustrationofthetaskdefinitionanddataprocessing.
methodologiesthatcandirectlyprobeforsuchintrinsicerrors.Our ? ?,? ?M?
andreplacingitscontentwitha[MASK]token,produc-
studyproposesanovelframeworktorigorouslyandautomatically ingacorruptedsentence???.
evaluateintrinsichallucinationinfinancialtabularreasoning. Theobjectiveofthemodel?
istorecovertheoriginalcontentof
themaskedspan? ?,?.Themodelisconditionedonthecorrupted
2.3 AutomaticDatasetConstruction sentenceandacontextset? ?:
Thehighcostandlimitedscalabilityofmanualannotationhave
???,? =?(??? ,? ?) (1)
driventhedevelopmentofautomatedevaluationdatasetconstruc-
tionmethods.Someapproachesachievefine-grainedanalysisby
decomposinggeneratedcontentintoatomicfactsforverification
Thecontext? ? ={T,P,? ??1 ,? ?+1 }includesthetables,theirpre-
text,andtheimmediatelyprecedingandsucceedingsentences.The
[1, 7, 13], while others leverage the inherent structure of rela-
goalisforthepredictedspan???,? tobematchedwiththeoriginal
tionaldatatoautomaticallygeneratecomplex,verifiablyquestion-
span?
?,?
answerpairs[18].Adversarialtestcasescanbedynamicallycreated
throughdataperturbation[26].AlthoughusingLLMsasannota-
3.2 MaskingCriteria
torsmayintroducebiases[28],recentworkshowsthatcarefully
designedautomatedpipelinescanyieldhigh-qualityannotations,
Toensurethetasksetupyieldsmeaningfulandunbiasedhallucina-
sometimesrivalinghumanperformance[12].Webuildonthese
tionevaluation,threekeyassumptionsmustholdforeachmasked
advancementsbyadaptingandextendingautomatedbenchmarking
span.Theseassumptionsguideourmaskingstrategyanddirectly
paradigmstotheuniquechallengesoffinancialtabularanalysis.
impactthereliabilityofevaluationoutcomes:
? Uniqueness:Themaskedspanmusthaveauniquecorrect
3 TaskDefinition
answer,preventingmultipleplausiblecompletions.
Inthissection,wefocusonthecorerequirementofretrievingin- ?
Example:Asentencelike"Thecompany?s[MASK]hasim-
formationfromcontextandprovidingaccurateanswersincomplex
proved"wouldbeexcluded,asseveralfinancialmetricssuch
financial scenarios, which is a common need for financial LLM
as"revenue,""profitmargin,"or"cashflow"couldallberea-
applications.Weformulatethetaskasacontext-awaremasked sonableanswers.
spanpredictionovertabularfinancialdata,forthegoalofintrinsic ? Consistency:Theground-
truthspanmustbeconsistentwith
hallucinationevaluation.Figure1demonstrateskeyflow.
thecontext,avoidinginternalmisalignmentswithinthesource
document.
3.1 ProblemFormulation ? Example:Ifthetablereportsoperatingincomeas$500million,
Letafinancialdocument?becomposedofasetofstructuredtables
themaskedsentenceshouldnotmistakenlystate"Operating
T andabodyoftext.Thistextispartitionedintotwodistinct,non- incomewas$500thousand"astheto-be-
maskedtruth.Evalu-
atingbasedontheincorrectmaskedcontentwouldunfairly
overlappingsets:
penalizeamodelthatcorrectlyalignswiththetable.
? AsetofexplanatorypretextsP,whereeachpre-text? ? ?P ?
Answerability:Themaskedspanmustbeinferablefromthe
introducesthetable? ?T.
providedcontext,ensuringthetaskissolvablebyanLLM.
? Asequenceofgeneralsentences? =(?
1
,?
2

,...,? ?),whichcon-
? Example:Ifthesentencestates"Thecompany?srevenuein-
stitutestheremainingtextofthedocument.
creasedby[MASK]in2024comparedto2023,"bothrevenue
Withinagivensentence? ? ??,weidentifyasetofnon-overlapping
figuresfor2024and2023shouldbepresentinthetablesor
spansofinterest,M? ={? ?,1 ,? ?,2 ,...,? ?,?},basedonspecificcri-
describedinthesurroundingtext;otherwise,thespanisnot
teria. A task instance is constructed by selecting a single span
answerableandshouldbeexcludedfromevaluation.
===PAGE 4===
ICAIF?25,November15?18,2025,Singapore,Singapore Zhang,etal.
Algorithm1Precision-RelaxedMatchingwithUnitGroups
(e.g.,percentageorabsolutevalue),therebyrestoringuniqueness.
Require: Groundtruthspan?,predictedspan??,unitgroupsU Tosolvethesecond,weintroduceaprecision-relaxedevaluation
Ensure: ??_???????_?????and??_????_????? protocolthatnormalizesandcomparesthenumericpredictions.
1: LetU scales ?Ubethesubsetofscaleunits Thisprotocolisdetailedin Algorithm1andconsistsoftwomain
2: (? ? ,? ?)?NormalizeNumber(?,U scales) components:
3: (? ?? ,? ??)?NormalizeNumber(??,U scales) Numeric Matching. This process first normalizes the ground-
//Normalizetobasevalueanddetermineprecision
truthspan?andthepredictedsp

15. assistant (2026-05-31T04:19:44.468Z)

This is rich. I now have the core empirical findings. Let me grab the results/discussion section (pages 5-8) where the quantitative breakdowns and key takeaways live, especially anything about VLMs and verification.

16. user (2026-05-31T04:19:49.653Z)

===PAGE 5 ===
FAITH:AFrameworkforAssessingIntrinsicTabularHallucinationsinFinance
ICAIF?25,November15?18,2025,Singapore,Singapore
Table1:AgreementamongLLMsandhumanannotationfor Table 2: Comparative statistics of the Pilot (human-
AnswerabilityAnnotation."Yes"and"No"represent"answer- annotated)andMain(LLM-annotated)datasetsplits.
able"and"unanswerable"respectively.TheLLMsusedare
GPT-4.1,Claude-sonnet-4,andGemini-2.5-pro.
Metric PilotSplit MainSplit
Numberofcompanies 9 453
Human
Avg.contextlength(chars) 14,148 12,843
LLMAgreementPattern Yes No Total Avg.numberoftables 14.9 19.2
Numberofsentences 164 1,122
All3Agree(Unanimous)
Numberofanswerablespans 300 2,406
3LLMs-No 1 626 627
3LLMs-Yes 276 11 287
2-1Split(Disagree)
2LLMs-Yes,1LLM-No 12 23 35 ResultsofOperations"(MD&A).Thissectionisselectedforitsrich
1LLM-Yes,2LLMs-No 11 164 175 narrativeanalysisandmanageablelength,providingadensesource
Total 300 824 1124 ofcontextualizedfinancialdata.Weapplyakeyword-basedsearch
tolocatetheMD&Asection,supplementingthisautomatedprocess
withmanualcurationtoensuredataquality.
DiscussionandAnalysis"sectionof10-Kreportsfromninecompa-
ContextandSentenceExtraction.Toavoidoverlapsbetween
niesacrossdiverseindustries(e.g.,healthcare,finance,technology).
contextandsentence,wefirstextracttablesandtheirimmediate
Each span is labeled for answerability by at least two financial
precedingsentenceastheexplanatorypre-texts.Thenweparsethe
experts,withaseniorrevieweradjudicatingdisagreements.This
restofthecontentintoplaintext.Afterwhichasentencesplitis

processyieldsahighinter-annotatoragreement(Fleiss?Kappa=
performedusingspaCytogetalistofcandidatesentenceswhich
0.905)andresultsin300answerableand824unanswerablespans.
incorporatedcustomrulestomergefragmentsthattypicallyarose
Theannotatorsalsoclassifythe300answerablespansintoourfour
asartifactsfromthedocumentconversionprocess.
financialreasoningscenarios(A:DirectLookup,B:Comparative,
NumericalSpanIdentification.Fromthissentencecorpus,
C:Bivariate,D:Multivariate),resultinginadistributionof212,58,
weisolateclaimscontainingnumericaldata.Thisisatwo-stage
22,and8respectively,whichreflectsthenaturalrarityofmore process:
complexreasoningtasks. ? InitialDetection:WeemployspaCy?sNamedEntityRecogni-
LLMAnnotation.WethenpromptthreeleadingLLMs(GPT-4.1,
tion(NER)todetectentitiessuchasMONEY,PERCENT,CAR-
Claude-Sonnet-4,Gemini-2.5-Pro)withtheanswerabilityannota-
DINAL,andQUANTITY.
tiontask.Theresults,summarizedinTable1,showthatunani- ?
SpanExpansion:Toensuretheextractedspansaresemanti-
mousLLMconsensusisanexceptionallystrongindicatorof
callycomplete,weexpandtheinitialNERoutputswithrulesto
answerabilitycorrectness.Whenallthreemodelsagreed,they
includerelevantcurrencysymbols(e.g.,$)andacomprehensive
correctlyidentified626of627unanswerablespans(99.8%accuracy)
vocabularyoffinancialunitsandscales(e.g.,"million,""billion,"
and276of287answerablespans(96.2%accuracy).
"pershare").
ThispilotstudyvalidatesthatleveragingunanimousLLMcon-
sensus is a highly reliable and scalable strategy for annotating LLM-based Answerability Annotation. To create a repre-
answerability.Itisworthnotingthatweonlymaskexistingtext
sentablesample,werandomlyselect10sentencesfromeach10-K
ratherthangeneratingnewcontent;thus,usingLLMsasannota-
report.Followingthesamesettinginpilotstudy,threepopular
torsdoesnotintroducefabrications.Thisensuresthesoundnessof
LLMsareemployedforannotatingwhethereachspanisanswer-
ourapproachforcreatingintrinsichallucinationevaluationbench- able given the provided context. A span is retained in the final
marks. datasetonlyifallthreemodelsunanimouslyclassifieditasanswer-
able.

4 EvaluationDataset

4.2 DatasetStatistics

4.1 DataCollectionandProcessing

Table2presentsastatisticaloverviewofourdataset,whichiscom-
Ourdatasetisbuiltuponpubliclyavailable10-KreportsfromS&P
posedofahuman-annotatedPilotSplitandalarger,LLM-annotated
500companiesfiledin2024,ensuringthedatareflectsreal-world
MainSplitcovering453S&P500companiesafterprocessingand
financialreportingpracticesandcoversawidespectrumofindus-
filtering.
tries.WesourcethesedocumentsinXBRLformatfromtheSEC?s
EDGARdatabase1.Thecollectedfilingsthenundergoseveralpro- A key characteristic of our dataset is its real

===PAGE 6 ===

Table3:ModelAccuracy(%)onPilotandMainSplits,includingscenarioperformance.Scenarioabbreviations:(A)DirectLookup,
(B)ComparativeCalculation,(C)BivariateCalculation,and(D)MultivariateCalculation.

Model PilotSplit MainSplit														
Overall Value Unit A B C D Overall Value Unit A B C D														
(n=212) (n=58) (n=22) (n=8) (n=1606) (n=635) (n=135) (n=10)														
Gemini-2.5-pro	95.0	95.7	97.7	96.7	91.4	95.5	75.0	91.9	97.8	93.1	91.8	94.0	96.3	90.0
Gemini-2.5-flash	91.0	92.3	96.7	94.8	82.8	90.9	50.0	88.7	91.1	93.7	90.2	88.0	87.4	70.0
Gemini-2.5-flash-lite	55.7	59.0	83.0	67.0	27.6	27.3	37.5	50.2	57.9	57.8	45.8	59.4	70.4	20.0
Claude-sonnet-4	93.0	93.7	97.3	96.2	93.1	81.8	37.5	95.6	95.8	98.7	97.0	82.6	94.1	80.0

Claude-haiku-3 64.7 66.0 92.7 71.7 50.0 59.1 0.0 81.3 81.8 93.9 84.1 80.6 66.7 30.0
GPT-4.1 89.7 90.3 95.3 93.9 87.9 72.7 37.5 89.2 89.5 92.6 91.1 90.4 77.8 30.0
GPT-4.1-mini 78.0 79.3 91.7 85.4 70.6 40.9 37.5 88.2 89.4 94.2 89.7 89.0 80.7 70.0
GPT-4.1-nano 31.3 43.3 62.3 36.8 19.0 22.7 0.0 70.0 71.9 92.1 71.8 72.3 53.3 10.0
Qwen-3-8B 20.3 32.0 44.3 24.5 10.3 13.6 0.0 30.6 35.6 36.1 27.1 35.7 54.1 0.0
Qwen-3-32B 68.0 70.0 96.0 80.7 34.5 45.5 37.5 73.9 76.7 83.7 73.5 76.1 81.5 40.0
Ministral-8B 22.3 23.0 75.7 29.7 1.7 9.1 12.5 40.8 41.8 74.7 45.3 33.2 31.9 0.0
Mistral-small-24B 63.7 63.7 97.7 77.8 29.3 31.8 25.0 45.6 64.2 53.4 45.3 46.0 57.0 0.0
Llama-3.1-8B 27.0 29.0 68.3 35.3 6.9 9.1 0.0 47.5 47.8 85.5 47.8 49.9 43.7 10.0
Llama-3.3-70B 56.0 57.0 95.3 66.0 36.2 31.8 0.0 37.0 40.9 42.5 34.7 40.9 53.3 10.0
Gemma-3-12B 32.0 34.3 83.7 42.5 6.9 9.1 0.0 15.2 19.7 37.7 16.3 12.8 17.0 0.0
Gemma-3-27B 37.3 39.0 91.0 45.7 20.7 13.6 0.0 33.8 37.3 51.0 31.7 38.6 43.7 0.0
5 Experiments 5.2.1 AQuantifiableHierarchyofModelReliability. Theresults
demonstrateadistinctperformancehierarchyamongtheevaluated
5.1 ExperimentalSetup
models.
Weevaluatearangeofstate-of-the-artopen-sourceandproprietary
Atoptierofproprietarymodels,ledbyClaude-Sonnet-4(95.6%
LLMsonourdataset?spilotandmainsplits.Duringevaluation,
onMainSplit)andGemini-2.5-Pro(91.9%onMainSplit),exhibits
wepromptthemodelstogenerateastep-by-steprationalebefore
highoverallaccuracy.Theirstrongperformance,whichremains
predictingthefinalmaskedvalueandtoself-classifytheirreasoning
largelyconsistentbetweenthePilotandMainsplits,validatesthe
processintooneofthefourfinancialscenarios(A-D).Thedetailed
robustnessofourbenchmark.However,the4-8%errorrate,while
promptisavailableinAppendixA.1.
lowrelativetoothermodels,remainsasignificantconsideration
Accuracyisusedastheprimaryevaluationmetric,computed
forfinancialapplicationswhereprecisionisnon-negotiable.
foroverallpredictionsandseparatelyforthenumericvalueand
Belowthistoptier,weobserveasecondgroupofcapablemod-
unitcomponents.Inaddition,wereporttheaccuracystratifiedby
els,includingGPT-4.1(89.2%)andGPT-4.1-mini(88.2%),which
financialscenario.Forthepilotsplit,scenariolabelsareassigned
deliverrespectablebutdemonstrablyloweraccuracy.Afurther
byhumanannotators.Forthemainsplit,weusethestrategystated
sharpdeclineisevidentinthemajorityofothermodels,particu-
insection3.4.Thisstratificationallowsforadetailedanalysisof
larlysmalleropen-sourcevariants.ModelssuchasLlama-3.1-8B
errortypes.
(47.5%)andQwen-3-8B(30.6%)exhibithigherrorfrequencies,ren-
Allmodelsarerunwithatemperatureof0toensurereproducibil-
deringthemunsuitablefortasksrequiringfinancialfidelity.
ity.ProprietarymodelsareaccessedviatheirofficialAPIs,while
open-sourcemodelsarerunon4H200GPUsusingtheLLaMA-
Factoryframework[29].Weimplementaretrymechanism(upto 5.2.2 Reasoning Complexity as the Primary
Performance Differ-
3attempts)tohandleinstanceswhereamodelfailedtoproducea entiator.
Theanalysisofperformanceacrossthefourreasoning
validJSONoutput. scenarios(A-D)revealsthattaskcomplexityisthemostsignificant
factorinfluencingmodelreliability.
DegradationinMulti-stepScenarios. Whilemostmodelsperform
5.2 ExperimentResults
adequately on Direct Lookup (A), accuracy systematically de-
This section presents a detailed analysis of the performance of
creasesastasksrequirecalculationandlogicalinference.Themost
variousLLMsonourfinancialhallucinationbenchmark.Thecom- pronouncedperformancedropocc

===PAGE 7 ===

Table4:KeyfinancialattributesforMohawkCommonsused Incontrast,GPT-4.1andClaude-
Sonnet-4failedtoutilizethe
inthemaskedvalueinferencetask.(Tableheadersandrows

debtinformationfromthetable.Instead,bothmodelsbasedtheir
havebeentruncatedforclarity.) predictionsolelyontheinformationinthesentence,bysimply
calculating90%ofthepurchaseprice(62.1M×0.90=55.9M).This
Investment Ownership% Debt($M) Rate Maturity
neglectsthemortgagedebtandindicatesinsufficientintegrationof
tabulardata.
Gotham 49% $8.5 8.36% Mar2024
ImplicationsforMultimodalReasoning: Thisexamplere-
MohawkCommons 18.1% $7.2 5.80% Mar2028
vealsakeylimitationincurrentLLMcapabilities:theinferenceof
Total $188.8 latentvariablesthatrequiresynthesizinginformationacrossmodal-
itiesandreasoningoverimplicitrelationships.OnlyGemini2.5
Proreconstructedthecorrectfinanciallogicchain,whereasother
modelsdefaultedtopredictingthemaskedspanwithoutgrounding
ResilienceinTop-TierModels. Incontrast,thetop-performing
answersintheprovidednumericaldata.
modelsdemonstratenotable,albeitimperfect,resilienceoncomplex
tasks.Gemini-2.5-ProisparticularlyrobustintheMultivariate
Calculation(D)scenario,achieving90.0%accuracyontheMain
6 ConclusionandFutureWork
Split.Claude-Sonnet-4alsoperformsstronglyat80.0%.While
theseresultsarepromisingandhighlightadvancedreasoningca-
Inthispaper,weintroducedanewdatasetandadynamicframe-
pabilities,a10-20%failurerateonthemostcomplexcalculations
workforevaluatinghallucinationsofLLMsinthefinancialdomain,
representsacriticalbarrier.Thisdifficultyinmaintainingfactual
providinganuancedviewoftheircurrentcapabilities.Ourfindings
consistencythroughmulti-steplogicremainsaprimaryvectorfor indicatethatevenstate-of-the-
artmodelsstrugglewiththenuances
intrinsichallucinations. offinancialtabulardata.Whileleadingmodelsareapproaching
theaccuracyrequiredforlesscriticalapplications,thefundamen-
5.2.3 CaseStudy. Beyondaggregatemetrics,weperformaquali-
talchallengesofensuringfactualintegrity,particularlyregarding
tativeanalysistodiagnosethemodels?specificfailuremodes.Our
numericalscaleincomplexreasoning,remaintheprimarybarrier
analysisrevealsseveralrecurringpatterns,withoneofthemost
totheirsafedeploymentinaccuracy-criticalfinancialworkflows.
significantbeingscaleerror.Thisoccurswhenamodelcorrectly
Forfuturework,weplantoexpandthebenchmarkingdatasetsto
identifiesanumericalvaluebutfailstoassociateitwiththeproper
includemoredocumenttypesandmorecomplexreasoningscenar-
magnitude(e.g.,reporting"$150"insteadof"$150million").Correct-
ios.Anotherpromisingdirectionistostudyhowfactorsliketable
ingforthissingleerrortypeinLlama-3.3-70B?soutputs,forinstance,
sizeandthenumberoftablesinthecontextaffecthallucination
wouldboostitsvalueaccuracyfrom37.0%to57.7%,highlighting
rates.Insummary,webelievethatrobust,domain-specificvalua-
thisasacriticalvulnerabilityincontextualnumericalgrounding.
tionisacrucialsteptowardsbuildingmorereliableandtrustworthy
However,moreprofoundfailuresstemfromadeficientunder-
LLMsforfinancialapplications.
standingoffinancialconceptsthatrequiremulti-stepreasoning
acrossdifferentdatasources.Toillustratethesedeeperchallenges,
wenowpresentarepresentativedetailedcasestudyonasample
7 Limitation
thatrequiredlatentvariableinferencefrombothtabularandtextual
context.Thetargetspanwastheequityinvestmentvalue($20.2 FAITH provides a systematic framework for assessing intrinsic
million)inthefollowingscenario: tabularhallucinationsinfinancialcontexts,yetseverallimitations remain.
OnJanuary27,2023,FundVacquireda90%interest
OurbenchmarkdrawsonpublicU.S.companyfilingsandcu-
inanunconsolidatedventurefor$20.2million,which
ratedfinancialtables.Thisdesignensuresdatareliabilityandcom-
purchasedashoppingcenter,MohawkCommons,lo-

parabilitybutlimitscoverageofdiversereportingstyles,regional
catedinSchenectady,NewYork,for$62.1million,in-
accounting standards, and sector-specific terminologies. Never-
clusiveoftransactioncosts.
theless,theframeworkisreadilyexpandabletoothercompanies,
Table 4 contains the relevant details for Mohawk Commons,
industries,andjurisdictions,providedthatstructuredfinancialdata
in

17. assistant (2026-05-31T04:20:01.521Z)

I now have the full picture of the empirical findings. I have everything I need: the quantitative model table, the "scale error" finding (highly relevant to muneem's transposed-magnitude concern), the degradation-with-complexity finding, and the specific open-source model failures. Let me extract the final claims.

Key findings relevant to the muneem research question:

1. **Scale errors** (correct digits, wrong magnitude ? e.g. "$150" vs "$150 million") are a named, recurring failure mode. Correcting this one error type lifted Llama-3.3-70B value accuracy from 37.0% to 57.7%. This directly bears on muneem's concern about transposed/wrong-magnitude numbers and validates magnitude-checking in verification.

2. **Even frontier models have 10-20% error on multi-step numerical reasoning**, and 4-8% overall error ? too high for "accuracy-critical financial workflows" without verification.

3. **Small open-source models fail catastrophically** on complex calculations (Qwen-3-8B 30.6%, Llama-3.1-8B 47.5%, Gemma-3-12B 15.2% overall on Main Split; near-zero on multivariate). This is highly relevant to the local-model recommendation ? small local LLMs are unreliable for financial number extraction/reasoning.

4. **Hallucination scales with task complexity** ? Direct Lookup is much more reliable than calculation. For muneem, pure extraction (lookup) is safer than asking a model to compute.

5. **Models defaulted to ungrounded predictions** rather than reading the table ? a hallucination pattern directly relevant to VLM-on-page-image risk.

Note one caveat for accuracy: this paper tests *text-based LLMs reading provided text/table context* (masked span prediction), not vision/OCR of page images. The "intrinsic hallucination on financial tabular reasoning" finding generalizes, but it is not an OCR/VLM-vision benchmark. I'll flag this in the claims framing.

18. user (2026-05-31T04:20:17.977Z)

Structured output provided successfully

19. assistant (2026-05-31T04:20:29.848Z)

I have completed the analysis. Summary of what I found and how I handled it:

Source: FAITH (ICAIF '25, National University of Singapore ? Asian Institute of Digital Finance). Peer-reviewed primary research, published in an ACM conference. High quality.

Fetch note: The initial WebFetch could not read the PDF (FlateDecode-compressed streams), returning only the abstract-level framing. I extracted the actual results by reading the locally-saved PDF with the project's `.venv` pdfplumber (global Python lacked it). This surfaced the quantitative model table and the discussion/case-study sections that contain the load-bearing claims.

Relevance caveat (important for synthesis): This paper benchmarks *text-based LLMs* on a masked-span-prediction task where the table/text context is *provided as text* ? it is **not** an OCR or vision-VLM-on-page-image benchmark. So it does not directly measure the OCR/layout error surface muneem cares about. What it does measure rigorously is *intrinsic numerical hallucination on financial tabular reasoning*, which generalizes to any model muneem feeds statement data to. The findings most useful to the research question:

1. Frontier models still hit 10-20% error on multi-step numerical reasoning ? never trust raw output; muneem's balance-reconciliation verification is the right posture.
2. Small open-source models (8B-24B: Qwen-3-8B 30.6%, Llama-3.1-8B 47.5%, Gemma-3-12B 15.2%) fail catastrophically ? a concrete caution against small/quantized local LLMs for financial-number work.
3. "Scale error" (right digits, wrong magnitude) is a top failure mode ? muneem should verify units/magnitude, not just digit sequences.
4. Error scales with reasoning complexity ? use the model as a pure extractor, not a calculator.
5. Models default to ungrounded predictions instead of reading the table ? the core hallucination risk for any page-image?JSON approach.

These are returned in the structured output. No files were written.